

RIFAUH HILAL SHABRIANSYAH

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PROFESSIONAL SUMMARY

Software Engineer holding an Applied Bachelor's degree (D4) in Informatics Engineering from State Polytechnic of Malang (GPA 3.60/4.00), focused on Web Back-End development (Laravel, FastAPI, Node.js) and IoT systems using ESP32/ESP8266. Experienced in building RESTful APIs, handling database integrations, writing microcontroller firmware, and automating workstation setups and Docker deployments. Completed internships at national shipyard PT PAL Indonesia (internal notification gateways) and digital IPTV broadcasting.

WORK & INTERNSHIP EXPERIENCE

PT. PAL INDONESIA – Surabaya, East Java, Indonesia

Software Engineer Intern (Back-End Developer)

Mar 2021 – Apr 2021

PT PAL Indonesia is Indonesia's state-owned shipyard specializing in the design, construction, and maintenance of naval vessels, commercial ships, and maritime engineering.

Key Contributions:

- Built backend modules for automated notification dispatch across WhatsApp, Telegram, and Email integrated with internal operational systems.
- Designed internal RESTful APIs for automated WhatsApp message delivery, speeding up alert distribution between divisions.
- Configured message queue processing to ensure reliable message delivery without server bottlenecks or timeouts during high-volume periods.

CV. Alva Indo Pratama – Malang, East Java, Indonesia

Android & Back-End Developer Intern

Jan 2020 – Mar 2020

Technology provider for Digital Television Broadcasting (IPTV, DVB-C/T2/S2), cross-platform applications (Android/Web/Desktop), and optical network infrastructure.

Key Contributions:

- Developed the EyePlus Android IPTV app for streaming digital television with responsive navigation and minimal buffering.
- Built backend web APIs to handle user authentication, channel playlist management, and real-time data sync.
- Added local client-side data caching to keep video playback smooth when user network signals fluctuated.

EDUCATION

State Polytechnic of Malang (Politeknik Negeri Malang) – Malang, East Java, Indonesia

Sep 2017 – Aug 2024

Bachelor of Applied Science (D4) in Informatics Engineering

GPA: 3.60 / 4.00

Core Curriculum: Software Engineering, Distributed Systems Architecture, Computer Networks, Embedded Systems & IoT, Relational & NoSQL Database Design.

TECHNICAL & PROFESSIONAL SKILLS

- **Soft Skills:** Analytical Thinking & Critical Problem Solving, Cross-Functional Team Collaboration & Technical Communication, High Initiative & Rapid Technical Adaptability, Time Management & Production Precision
- **Programming Languages:** Kotlin, Python, PHP, JavaScript / TypeScript, C++, Java, SQL
- **Frameworks & Back-End:** Laravel (v10 - v12), FastAPI / Flask, React (Inertia.js), Node.js / Express, RESTful API & Microservices, Tailwind CSS

- **IoT & Embedded Systems:** ESP32 / ESP32-CAM, ESP8266 (Wemos D1), Arduino C++, Telemetry Sensors & Actuators, MQTT & HTTP REST Protocols
- **Tools, DevOps & Databases:** Docker & Podman Containerization, Git & GitHub Actions (CI/CD), MySQL, PostgreSQL, MariaDB, MongoDB & Firebase, Antigravity CLI & MCP, Android Studio
- **Operating Systems:** Linux (Ubuntu Server, Debian, Fedora), Windows 10 / 11, Remote Systems Administration (SSH, RDP, VNC)
- **Languages:** English (Professional Working Proficiency), Indonesian (Native Proficiency)

FEATURED PROJECTS & GITHUB PORTFOLIO

SmartAttend - Face Recognition & RFID Attendance System

2024

(<https://github.com/Hilal06/smart-attendanceESP32>)

Tech Stack: Laravel 12, React 19 (Inertia.js V2), Python (Face Recognition), Docker, ESP32-CAM, RFID

Automated attendance system combining facial recognition (ESP32-CAM & Python) and RFID cards, connected directly to a Laravel 12 and React 19 administrative web dashboard. Built for practical school use to eliminate morning entrance queues.

- Built an administrative school web dashboard using Laravel 12 and React 19 (Inertia.js V2) to monitor daily attendance records and student statuses in real-time.
- Developed a Python face recognition service utilizing the face_recognition library with verification latency under 500ms.
- Connected the ESP32-CAM microcontroller and RC522 RFID reader to the web server via Bearer token-authenticated REST APIs.
- Packaged the full application stack into Docker containers to simplify school server installation and maintenance (SMK Negeri 2 Kraksaan & SMP Negeri 1 Pakuniran).

win-fresh-setup - Windows 10/11 Automated Application Installer

2026

(<https://github.com/Hilal06/win-fresh-setup>)

Tech Stack: Python 3, Winget Package Manager, Rich TUI, PowerShell, Automation CLI

Interactive Python terminal tool (TUI) to automate daily software and developer tool installations on fresh Windows 10/11 setups via Windows Package Manager (Winget).

- Built an interactive terminal UI using the Rich library with category menus, arrow key navigation, and real-time installation progress bars.
- Automated batch installation of essential developer tools (VS Code, Git, Docker, Node.js) in minutes without manual installer downloads.
- Added automatic error handling to let batch installations continue smoothly even if a specific package download fails.

personal-fresh-linux-setup - Modular Linux Post-Installation Suite

2026

(<https://github.com/Hilal06/personal-fresh-linux-setup>)

Tech Stack: Shell / Bash, Fedora & Ubuntu, Zsh + Starship, Hardware Tuning, TUI Interface

Modular Bash script collection for rapid post-installation setup on Linux distros (Fedora & Ubuntu), covering dev tools, Zsh shell, terminal prompts, and hardware tuning.

- Provided a bilingual (ID/EN) terminal interface to select installation components including dev toolchains, browsers, and ASUS TUF/ROG laptop configurations.
- Automated Zsh terminal setup with Starship prompt and organized dotfiles, complete with automatic backup for existing configs.
- Reduced Linux machine setup time from several hours to around 15 minutes through clean modular scripts.

Ethereal Wedding Invitation - Serverless Digital Event Platform

2026

(<https://github.com/Hilal06/ethereal-wedding-invitation>)

Tech Stack: React, TypeScript, Vite, Tailwind CSS, Google Apps Script, Google Sheets API

Digital wedding invitation website built with React and Tailwind CSS featuring an RSVP form and guestbook that saves directly to Google Sheets via Google Apps Script without backend hosting costs.

- Built a responsive, lightweight single-page application with React, TypeScript, and Tailwind CSS loading in under 1.5 seconds on mobile phones.
- Used Google Apps Script as a lightweight API bridge to read and write guest RSVP records directly to Google Sheets with zero server costs.
- Implemented frontend form validation and spam protection to prevent duplicate or empty guestbook submissions.

Wed Planner - Modern Wedding Preparation & Budget Manager

2026

(<https://github.com/Hilal06/Wed-Planner>)

Tech Stack: Kotlin, Jetpack Compose, Clean Architecture, Room DB, Cloud Sync, Home Screen Widgets

Android application to help couples organize wedding preparations, from budget tracking and priority to-do lists to interactive home screen widgets.

- Implemented Clean Architecture and MVVM pattern in Kotlin and Jetpack Compose for a clean, modular, and easily testable codebase.
- Built a practical budget tracking feature with categorized cost breakdowns and automatic alerts when approaching budget limits.
- Integrated device-to-device cloud sync between partners and added Glance Home Screen Widgets for quick task updates right from the home screen.

LabelStudioBot - Python Project (<https://github.com/Hilal06/LabelStudioBot>)

2026

Tech Stack: Python

Software application built with Python for Python workflows, featuring a clean modular codebase.

- Developed core functional modules using Python focusing on execution reliability and clean data flow.
- Organized project structure with straightforward error handling and maintainable code separation.
- Maintained version control history and technical setup documentation to simplify local environment reproduction.